Lecture 8: Numerical optimization and integration

ECON 7510

Cornell University

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Numerical optimization

Motivating example: Logit model of demand

Suppose an employer offers its employees two health plans, $a = 0, 1$. The cost of both is the same, but they differ in the specifics of their coverage.

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- So, conditional on x, the share of employees who choose $a = 1$ is

$$\begin{aligned} p(x) &\equiv \Pr(a_i = 1 \mid x_i = x) \\ &= \Lambda(x'\beta) \\ &= \frac{\exp(x'\beta)}{1 + \exp(x'\beta)} \end{aligned}$$

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Estimation problem

- Suppose we observe (x_i, a_i) for $i = 1, \ldots, I$.
- Estimate by maximum likelihood:

$$\begin{aligned}\hat{\beta}_{\text{MLE}} &= \text{argmax}_\beta \prod_i p(x_i;\beta)^{a_i}\,(1-p(x_i;\beta))^{1-a_i} \\ &= \text{argmax}_\beta \sum_i \underbrace{a_i \log p(x_i;\beta) + (1-a_i)\log(1-p(x_i;\beta))}_{\ell(x_i;\beta)}\end{aligned}$$

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$$\begin{aligned}\nabla_\beta \ell(x_i;\beta) &= \left[\frac{a_i}{p(x_i;\beta)} - \frac{1-a_i}{1-p(x_i;\beta)}\right] \nabla_\beta p(x_i;\beta) \\ &= \frac{a_i - p(x_i;\beta)}{p(x_i;\beta)\left(1-p(x_i;\beta)\right)} \nabla_\beta p(x_i;\beta)\end{aligned}$$

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- Gradient of the objective function:

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- **Question**: What is a geometric interpretation of the gradient that might be useful when thinking about optimization?

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- **Question**: What is a geometric interpretation of the gradient that might be useful when thinking about optimization? The gradient points in the direction of (locally) steepest ascent.

→ Key idea: if start at some β and want to move toward $\hat{\beta}_{\text{MLE}}$, the gradient is a good direction to go.

Gradient Descent: A simple algorithm for numerical optimization

$$\min_{\theta} f(\theta)$$

Algorithm:

1. Start at some θ_1.
2. Let $\theta_{k+1} = \theta_k - \eta \nabla f(\theta_k)$.
3. Repeat step 2 until $\nabla f(\theta_k) \approx 0$.

η is the *learning rate*. Gradient may give a good direction locally, but a bad direction more globally.

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Applying gradient descent

See `L8.py`

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- What about **constrained** optimization problems?

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$$\min_{\theta} f(\theta) \text{ subject to } g(\theta) \leqslant 0$$

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- As it turns out, what works much better in practice is **Augmented Lagrangian** method.

Numerical constrained optimization

Augmented Lagrangian method

$$\mathcal{L}(\theta, \lambda, \kappa) = f(\theta) + \lambda' g(\theta) + \frac{\kappa}{2} g(\theta)' g(\theta)$$

- Choose initial parameters θ_0
- While not converged:
  - Primal update: $\theta \leftarrow \arg\min_\theta \mathcal{L}(\theta, \lambda, \kappa)$
  - Dual update: $\lambda \leftarrow \lambda + \kappa g(\theta)$

κ is a hyperparameter that has to be chosen in advance.

Numerical constrained optimization

A lazier (and better) method!

- `torch` and `scipy.optimize` work well together, and the latter can handle constrained problems.
- See `scipy-constrained-optimization-pytorch-example.py` for an example implementation.

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$$\bar{\epsilon} = \mathbb{E}\left[\frac{\partial p(x)}{\partial x}\frac{x}{p(x)}\right] = \int \frac{\partial p(x)}{\partial x}\frac{x}{p(x)}\phi(x)dx$$

Numerical integration

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$$\begin{aligned}\frac{\partial p(x)}{\partial x} &= \frac{\partial}{\partial x}\frac{\exp(x'\beta)}{1+\exp(x'\beta)} \\ &= p(x)\left(1-p(x)\right)\beta\end{aligned}$$

$$\begin{aligned}\epsilon(x) &= \frac{\partial p(x)}{\partial x}\frac{x}{p(x)} \\ &= \left(1-p(x)\right)\beta x\end{aligned}$$

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2. Gauss-Hermite quadrature: Exploit a special property of the Normal distribution.

Gauss-Hermite quadrature

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$$\begin{aligned}\mathbb{E}\left[h(x)\right] &= \int_{-\infty}^{\infty} h(x)\phi\left(\frac{x-\mu}{\sigma}\right)dx \\ &\approx \frac{1}{\sqrt{\pi}}\sum_{j} w_j h(\sqrt{2}\sigma x_j + \mu)\end{aligned}$$

If we choose $w_1, \ldots, w_n$ and $x_1, \ldots, x_n$ wisely, we can show that this approximation is very good

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If we choose $w_1, \ldots, w_n$ and $x_1, \ldots, x_n$ wisely, we can show that this approximation is very good; in fact, it is *exact* if h is a polynomial of degree $2n-1$ or less.

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More general case

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Suppose $x \sim F$ and $z \sim \Phi$. Note that $F(x), \Phi(z)$ are both distributed $U[0,1]$. It follows that

$$F^{-1}(\Phi(z)) \sim F$$

has the same distribution as x.

Gauss-Hermite quadrature

More general case

Change of variable:

$$
\begin{aligned}
\mathbb{E}\left[h(x)\right] &= \int h(x) f(x) dx \\
&= \int h\left(F^{-1}(\Phi(z))\right) f(F^{-1}(\Phi(z))) \frac{dx}{dz} dz \\
&= \int h\left(F^{-1}(\Phi(z))\right) \phi(z) dz \\
&= \mathbb{E}\left[h\left(F^{-1}(\Phi(z))\right)\right] \\
&\approx \frac{1}{\sqrt{\pi}} \sum_j w_j h\left(F^{-1}\left(\Phi\left(\sqrt{2} z_j\right)\right)\right)
\end{aligned}
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(The third step uses the Inverse Function Theorem.)

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Question: When doing this kind of change of variable, how should that affect our choice of the number of quadrature points?